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REVIEW ARTICLE



Innovative approaches in acute graft-versus-host disease: emerging therapeutics and novel clinical trial design

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ABSTRACT

Acute graft-versus host disease (aGVHD) is a major early complication of allogeneic hematopoietic cell transplantation and is associated with substantial morbidity and mortality, particularly in severe cases. The use of corticosteroids as first-line therapy remains standard of care, despite limited efficacy in high-risk cases and associated toxicities. Ruxolitinib has emerged as the treatment of choice for patients who fail corticosteroids, but there remains a need for novel treatment strategies. In this review, we examine the evolving therapeutic landscape of aGVHD and summarize emerging approaches including small molecule inhibitors, biologics, and cellular- and microbiome-modulating therapies. We explore how improved understanding of aGVHD pathophysiology continues to inform new interventions while also addressing the operational and conceptual challenges that constrain clinical trial design and regulatory progress. Finally, we discuss new innovations in biomarker-driven strategies and adaptive clinical trial designs that may facilitate future therapeutic development and advance outcomes for patients with aGVHD.

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Introduction

Acute graft-versus-host disease (aGVHD) is a major complication of allogeneic hematopoietic cell transplantation (HCT), occurring when donor immune cells target and damage recipient tissue. Classically targeting epithelial tissues, including the skin and gastrointestinal (GI) tract, as well as the liver, both presentation and disease severity are highly variable, ranging from mild, self-limiting symptoms to fulminant multi-organ dysfunction [1,2]. Up to 40%–50% of transplant recipients experience symptoms related to aGVHD, establishing it as one of the main causes of morbidity after HCT [3]. Severe aGVHD (grades III–IV), particularly involving the lower GI tract, is a major contributor to non-relapse mortality (NRM) in the first year after HCT [4]. Notably, aGVHD typically arises primarily in the early post-transplant period and is both clinically and biologically distinct from chronic GVHD [5].

In this review, we will provide a comprehensive overview of the current therapeutic landscape of aGVHD, explore emerging investigational therapies, and critically examine ongoing challenges in therapeutic innovation and trial development.

Historical approaches in aGVHD treatment

An understanding of the broader historical landscape of aGVHD therapy is key to contextualizing recent and ongoing advancements in the field. For clinically significant aGVHD (grades II–IV), systemic therapy with high-dose systemic corticosteroids remains the cornerstone of first-line treatment. Although often effective in mild disease, patients with severe aGVHD are less likely to exhibit a meaningful response to steroid therapy than those with less severe manifestations [6]. Furthermore, prolonged therapy with systemic corticosteroids is associated with significant toxicities including hyperglycemia, hypertension, muscle wasting, osteoporosis, and increased infection risk. Thus, between 35 and 50% of patients with aGVHD are estimated to require second-line therapy due to lack of adequate response to corticosteroids, known as steroid-refractory aGVHD (SR-aGVHD) [6,7].

Numerous additional agents and modalities have been investigated over the past decades, including antithymocyte globulin (ATG), tumor necrosis factor (TNF) antagonists, interleukin-2 (IL-2) modulation, as well as traditional immunosuppressive agents such as mycophenolate mofetil (MMF). However, none of these

has achieved widespread adoption due to inconsistent efficacy or unfavorable toxicity profiles. Prior to its inclusion in prophylactic regimens for GVHD, ATG was evaluated in combination with corticosteroids in initial management of aGVHD and found to confer an increased infection risk and no clear survival benefit [8]. TNF antagonists, such as etanercept, have shown higher complete response rates (CRR) compared to steroids alone yet carries significant infectious risk [9,10]. Efficacy of IL-2 receptor antagonists varies by agent and disease context; although this class of therapeutics demonstrate moderate efficacy, infection risk and lack of large controlled trials limit their current usage [11,12]. MMF has a well-established role in prophylactic regimens but is available as a treatment for aGVHD due to severe infection risk [10,13].

Extracorporeal photopheresis (ECP) has been employed as a steroid-sparing, second-line or salvage therapy with some evidence to support its clinical efficacy and safety profile. ECP is unique in that it is not a directly immunosuppressive treatment; by exposing patient mononuclear cells, isolated by leukapheresis, to ultraviolet light along with 8-methoxypsoralen prior to reinfusion, it appears to induce immune tolerance mechanisms, potentially through regulatory T cell induction. However, its exact mechanism of action remains unclear [14]. One meta-analysis of prospective studies investigating ECP in the aGVHD setting found a pooled overall response rate (ORR) of 69%, with an ORR of 65% for gastrointestinal involvement, with no adverse events in any aGVHD patients [15]. However, heterogeneity in endpoint definition across these studies as well as differences in ECP protocols at centers suggests a need for larger-scale, rigorous trials that can truly delineate the efficacy of this treatment strategy.

Existing and emerging therapeutic approaches for aGVHD

In recent years, a growing array of therapeutic strategies has emerged for the treatment of aGVHD. These novel agents are intended to modulate the underlying immunopathology of aGVHD while mitigating the toxicities of broad immunosuppression. Ranging in class from small molecule inhibitors and biologic agents to cell-based therapies and microbiome-modulating strategies, these approaches leverage current understanding of aGVHD pathophysiology on its most fundamental level.

Ruxolitinib

Ruxolitinib, a selective Janus Kinase (JAK) 1/2 inhibitor, is currently the only second-line therapy with broad

international consensus support [16,17]. Its introduction marked a paradigm shift for the treatment of SR-aGVHD. Inhibition of the JAK/STAT signaling pathway reduces pro-inflammatory cytokine production and T-cell activation, targeting key drivers of aGVHD pathogenesis [18]. The pivotal phase 3 REACH2 trial demonstrated ruxolitinib (10mg oral, twice daily) achieve an ORR of 62% (CR 34%) in SR-aGVHD, significantly outperforming the best available therapy as chosen by the investigators (ORR 39%). Durable ORR at day 56 was assessed at 40% with ruxolitinib, compared to 22% with control [19]. Subsequent meta-analyses confirmed pooled ORRs in the 65-75% range for SR-aGVHD, and REACH4, which sought to evaluate ruxolitinib in the pediatric setting, reported a day 28 ORR of 84.4% and CR in up to 56% of patients [20,21]. Based on these data, ruxolitinib is considered the standard of care for SR-aGVHD and remains the only FDA-approved therapy to date for SR-aGVHD in adults. Its consistent efficacy across age groups and degree of organ involvement underscores the central role it plays in current aGVHD management [19,22,23]. However, despite an array of third-line options, no consensus exists beyond ruxolitinib (Table 1), highlighting the need for more effective and organ-specific therapies [16,17,24].

Additional small molecules

Building off the success of ruxolitinib, the selective JAK1 inhibitor itacitinib was developed with the rationale that JAK1-selective inhibition might reduce inflammatory cytokine signaling while avoiding the myelosuppression and cytopenias that can occur with ruxolitinib. The phase 3 GRAVITAS-301 trial randomized participants with grades II-IV aGVHD to receive itacitinib or placebo in combination with

Table 1. Suggested systemic agents for steroid-refractory acute GVHD, according to the National Comprehensive Cancer Network guidelines.

FDA-approved category 1 agents

Ruxolitinib

Alternative agents (alphabetical order)

Alemtuzumab

Alpha-1 antitrypsin

Anti-thymocyte globulin

Basiliximab

Calcineurin inhibitors

Etanercept

Extracorporeal photopheresis

Infliximab

mTOR inhibitors

Mycophenolate mofetil

Pentostatin

Tocilizumab

Urinary-derived human chorionic gonadotropin/epidermal growth factor

Vedolizumab

corticosteroids as first-line therapy for aGVHD. The use of itacitinib did not significantly improve day 28 ORR (itacitinib, 74% vs placebo, 66%; OR 1.45, $p=0.078$) or NRM. However, day 28 CR rates were higher with itacitinib, and a greater number of participants in the placebo group discontinued treatment due to aGVHD progression or inadequate response [25]. In a single-arm phase 2 trial, itacitinib was administered to participants with low-risk aGVHD. Compared to matched control patients receiving corticosteroids, participants receiving itacitinib achieved similar day 28 ORR (itacitinib, 89% vs steroids, 86%; $p=0.67$), although with more responses within the first 7 days of treatment and fewer serious infections [26]. No head-to-head comparisons of itacitinib and ruxolitinib have been conducted.

Alpha-1-antitrypsin (AAT) is a serine protease inhibitor with anti-inflammatory and immunomodulatory effects that has emerged as a novel immunomodulatory approach in aGVHD [27]. AAT was initially evaluated in two separate trials as monotherapy in the context of SR-aGVHD, in which the ORR was as high as 65%, with included CRs and responses in all GVHD target organs [28,29]. The most recent and pivotal evidence to support the use of AAT in the management of aGVHD is the preliminary results from a multicenter phase 3 trial (BMT CTN 1705), which compared AAT to placebo in combination with standard corticosteroid treatment as first-line therapy for high-risk aGVHD. In the initial intention-to-treat analysis, the primary endpoint (Day 28 ORR) was met for 60% for participants assigned to AAT and 45% for participants assigned to placebo. There were also more complete responses achieved in participants receiving AAT than placebo (55% vs 45%) [30]. This has sparked a renewed interest in AAT as a therapeutic option in aGVHD and future studies should seek to clarify the optimal positioning of AAT in the therapeutic landscape.

Biologic therapies

Apraglutide, a long-acting glucagon-like peptide-2 (GLP-2) analog, promotes intestinal mucosal repair and barrier function by activating GLP-2 receptors on epithelial cells, which stimulate enterocyte proliferation and regeneration, restoring cell populations depleted in lower-GI aGVHD [31]. The STARGAZE trial was a multicenter, single arm phase 2 trial evaluating the safety, feasibility and clinical efficacy of adding apraglutide to ruxolitinib for the treatment of steroid-refractory lower-GI aGVHD. Apraglutide was administered weekly as a subcutaneous injection and was well tolerated. On Day 28, the lower GI ORR and CR rate were 54.8%

and 29%, respectively [32]. A subsequent analysis compared the interim results from the STARGAZE trial with a cohort of patients receiving ruxolitinib alone for lower-GI aGVHD constructed from the Mount Sinai Acute Graft-Versus-Host Disease International Consortium (MAGIC) observational trial data. A higher proportion of trial participants treated with apraglutide and ruxolitinib achieved ORR at Day 28 than patients treated with ruxolitinib alone (58.1% vs 38.8%), with trends in CR rates and NRM both favoring the use of apraglutide [33]. Additionally, two reports have reported clinical responses with teduglutide, another GLP-2 analog, further supporting the role of this class of agents in the treatment of lower-GI aGVHD [34,35].

Vedolizumab is a humanized monoclonal antibody which specifically targets $\alpha 4 \beta 7$ integrin, which is expressed on T lymphocytes and interacts with mucosal addressin cell adhesion molecule-1 (MAdCAM-1), which found on gut-associated endothelial cells, thus directing T-cell trafficking to the GI tract; by binding this surface receptor, vedolizumab prevents migration and infiltration of T cells to the GI mucosa, thus attenuating GI tract inflammation and tissue damage in lower-GI aGVHD [36]. A randomized, placebo-controlled phase 3 trial demonstrated that vedolizumab in combination with calcineurin inhibitors reduced the incidence of lower GI aGVHD and improved 6-month GI GVHD-free survival [37,38]. However, the evidence is less robust for its role in aGVHD treatment [39,40]. Meta-analysis of vedolizumab in SR lower-GI aGVHD found a pooled ORR of 50% at day 28; 6- and 12-month overall survival (OS) rates from this analysis were low at 34.5% and 29.2% [41]. However, these patients represented a particularly morbid subset of aGVHD patients with high-risk GI involvement prior to inclusion in the trial. Natalizumab, a related $\alpha 4$ integrin blocker, has been studied in two phase 2 trials as a corticosteroid adjunct but was not found to improve OR or CR compared to steroids alone (OR 60% vs 58%), or lend a survival benefit [42,43]. The totality of data suggest that this class of agent may be more effective as a preventative intervention than as a therapy for disease.

Itolizumab is a humanized monoclonal antibody which binds to and blocks the function of CD6, a co-stimulatory molecule expressed on T cells which interacts with ALCAM (CD166) on antigen-presenting cells and tissue, and has a key function in T-cell activation, proliferation, and trafficking [44]. Although an early phase 1 clinical trial for itolizumab in combination with corticosteroids as first-line therapy in newly diagnosed Grade III-IV aGVHD demonstrated a day 29 ORR of 68%, this result has not yet been recapitulated

in studies with larger sample sizes [45]. The Phase 3 EQUATOR trial has, to date, showed no statistically significant improvement in ORR at day 29, though statistically significant improvements were noted in numerous secondary endpoints such as durable CR rate at day 99 and failure-free survival [46].

Tocilizumab is a humanized monoclonal antibody that blocks the interleukin-6 (IL-6) receptor. Its proposed mechanism of action in the context of aGVHD is suppression of IL-6 mediated pro-inflammatory signaling, which is classically implicated in T-cell activation and recruitment as well as Th17 differentiation. IL-6 blockade limits effector T-cell expansion and attenuates the cytokine-driven inflammation characteristic of lower GI aGVHD [47]. Regarding efficacy, small single-center retrospective studies have reported a complete response of 62.5% at 11 days following tocilizumab initiation in patients with severe steroid-refractory lower-GI aGVHD; however, significant safety concerns relating to infectious complications highlight the risk of immunosuppression with IL-6 blockade. One center reported bacterial infections in up to 69% of patients receiving tocilizumab. While mechanistically attractive, its use in the treatment setting remains investigational and is not widely adopted [48,49].

Cellular therapies

Mesenchymal stromal cells (MSCs) and decidual stromal cells (DSCs) are under active investigation for their immunomodulatory properties. Remestemcel-L is a bone marrow-derived MSC product, currently approved for pediatric SR-aGVHD in the US. MSCs inhibit T cell activation and thus reduce both proliferation and production of pro-inflammatory cytokines. Pharmacodynamic studies support this as well, finding that remestemcel-L administration in pediatric patients with SR-aGVHD at 2×10^6 cells/kg led to reductions in inflammatory biomarkers – tumor necrosis factor receptor 1 (TNFR1) decreased by an average of 79% at day 180 – as well as circulating activated T cells – the CD3+CD4+CD25+HLA-DR+ population had decreased by 64% by day 180 – in the same population. Clinical evidence from this same pivotal MSB-GVHD001 trial demonstrated a day 28 ORR of 70.4% with durable response rates exceeding 65% at day 180. Collectively, these data established remestemcel-L as an effective and durable option for pediatric SR-aGVHD, supportive of FDA approval for children [50]. Additional trials will be needed to elucidate the role of remestemcel-L in the current treatment algorithm for SR-aGVHD in adults.

DSCs, derived from placental tissue, represent an additional immunomodulatory cell therapy strateg

in aGVHD. DSCs are thought to exhibit enhanced immunosuppressive properties relative to MSCs, likely reflecting their physiologic role in maintaining maternal-fetal tolerance. DSCs suppress T cell proliferation and promote regulatory T cell expansion, thus opposing the pro-inflammatory milieu of aGVHD pathogenesis. Clinical evidence from the Karolinska Institute found that DSCs were associated with improved survival in SR-aGVHD. In severe SR-aGVHD, patients treated with DSCs plus 5% albumin achieved a one-year survival of 73%, compared with 3% in historical controls and 20% in a parallel BM-MSC study [51,52]. DSCs are a promising and mechanistically distinct cellular therapy approach to aGVHD, with early data supporting robust improvements in survival.

Fecal microbiota transplantation

Given the strong link between intestinal microbiota diversity and GVHD severity, fecal microbiota transplant (FMT) has emerged as an experimental treatment modality for lower GI aGVHD [53]. All aGVHD target organs are exposed to microbial antigens, and dysbiosis correlates with increased GVHD severity [54]. Multiple early-phase trials and institutional reports have established that FMT can result in clinically meaningful responses in lower-GI aGVHD, even after failing multiple lines of therapy [55,56]. Additional investigation efforts include the single-arm phase 2a HERACLES trial, which evaluated the efficacy of pooled allogeneic fecal microbiota (MaaT013) in patients with grade III-IV SR lower aGVHD. GI ORR reached 58% in patients treated through an expanded access program [57]. Finally, an initial single-arm pilot study of primarily treatment-naïve patients with high-risk lower GI aGVHD found a complete response of 70% at day 28, suggesting that FMT may have strong potential not just as salvage therapy for steroid-refractory lower GI aGVHD, but as a key first-line therapy for in aGVHD with lower GI involvement [58].

Unmet clinical needs

Despite the progress made in recent years, the treatment of aGVHD remains fraught with substantial limitations. Although several agents have been introduced or repurposed for use in steroid-refractory disease, innovation has not kept pace with clinical need—mainly, patients with high-risk features, refractory disease beyond second-line therapy, or non-intestinal phenotypes. This section outlines key barriers that continue to constrain the development and effective deployment of novel aGVHD treatments.

Effective therapies for severe and high-risk disease

While advanced strategies for aGVHD prophylaxis have reduced the incidence of severe aGVHD after allo-HCT, the prognosis for patients who do go on to develop grades III-IV or biomarker-defined high-risk aGVHD remains dismal with current therapies [59]. In the REACH1 trial investigating ruxolitinib for steroid-refractory aGVHD, ORR in Grades III-IV disease hovered around 40%, compared to the grade II subgroup in which 82.6% of patients had some degree of response. In the same trial, the presence of stage $\geq$ III lower GI involvement predicted a significant reduction in OS [22]. Additionally, a biomarker-based risk stratification of aGVHD patients found that while ruxolitinib outperformed other therapeutic approaches in low-risk patients, those with high-risk disease had dismal outcomes regardless of treatment choice. Patients with high-risk disease were unlikely to achieve CR, and the patients who lacked a CR by Day 28 (regardless of partial responses vs no response) had similarly high 6-month NRM [23]. These studies illustrate current limitations of the therapeutic landscape for treatment of grades III-IV steroid-refractory aGVHD and the need for treatments that can achieve timely and deep responses.

Lack of tissue specificity in current interventions

Another challenge is organ-specific efficacy. Many recent novel agents have mechanisms of action which are limited to lower GI disease. While this has helped address the more guarded prognosis of lower GI aGVHD, it leaves cutaneous and hepatic aGVHD that fails ruxolitinib comparatively underserved, despite skin being the most common and often earliest presentation of disease. While non-life threatening, there is a significant burden to treating skin disease with corticosteroids, and disease that persists despite ruxolitinib therapy can be quite challenging. We acknowledge that recent investigation into AAT and itolizumab, respectively, could potentially help expand therapeutic options for patients with non-GI disease.

Development of third-line therapies and clinical trial design

Therapeutic options beyond second-line treatment are remarkably limited, and many agents used in the third-line setting are investigational, poorly studied, or used off-label based on limited retrospective data. Robust clinical trial development in this setting is

challenged by small, heterogeneous patient populations with poor performance status and rapid clinical decline. Randomized controlled trials are not feasible; instead, most available data come from small single-arm phase 1/2 studies or expanded access programs [50,57]. Furthermore, patients with hepatic GVHD or other severe complications such as ongoing infection are often excluded despite being the population most in need of such interventions, limiting generalizability and leaving critical therapeutic gaps [22,57]. While these patients may be of the highest need, the concern is that investigational agents will fail to demonstrate their true clinically efficacy signal in this population.

Timing of therapeutic interventions

An emerging consensus suggests that earlier intervention may be critical to improving outcomes in aGVHD. Many pathologic features of GVHD—particularly epithelial barrier loss and crypt dropout—can be irreversible once established. It is therefore reasonable to expect that agents may show greater efficacy when administered before extensive tissue injury has occurred, but regulatory frameworks and traditional trial designs have delayed their use until after failure of corticosteroids; one report found that beginning ruxolitinib within 7 days of corticosteroid failure significantly improved complete response rates (69% CR in the <7 day cohort vs. 25% if started after 7 days) [60]. A number of trials are now investigating combination approaches at initial diagnosis, such as the previously mentioned BMT CTN 1705 study of AAT plus corticosteroids [61].

Advancing aGVHD therapeutics through innovations in clinical trial design

The development of new therapies for aGVHD is complicated by evolving challenges in trial design, in part due to advances in prophylaxis and standard of care treatment of aGVHD. While such therapeutic innovations as post-transplant cyclophosphamide prophylaxis have transformed patient care, they have paradoxically made it more difficult to study severe aGVHD by dramatically reducing its incidence. By narrowing the pool of eligible patients for clinical trials of agents in severe aGVHD, large phase 2 or 3 studies are made increasingly impractical due to cost, duration, and enrollment challenges. Similarly, the establishment of ruxolitinib as the standard of care for SR-aGVHD has redefined trial design as it becomes difficult to justify control

arms that exclude it. We henceforth highlight approaches to be considered when developing therapeutics in the current treatment landscape of aGVHD (summarized in Figure 1).

Use of single-arm trials with historical or matched controls

As illustrated above, therapeutic innovation in aGVHD is currently impaired by challenges such as small patient populations despite the urgent need for new, effective therapies. Increasing employment of single-arm trials with historical or contemporaneous matched controls helps address this obstacle, as exemplified by the phase 2 trial of remestemcel-L in pediatric groups, where a single-arm trial demonstrated high and durable day 28 ORR compared to historical benchmarks [50]. This approach has been used in additional phase 2 trials, particularly those from the MAGIC consortium [62,63]. The use of well-characterized historical controls expedites trial completion and therefore access to promising therapies for patients with high unmet need. Such designs are best used in similarly rare and high-mortality conditions as subgroups of aGVHD; randomized control cohorts remain the gold standard for safety and efficacy and single-arm trials should be regarded as a pragmatic and complementary approach rather than a replacement.

Improved trial precision through biomarker-based risk stratification

The incorporation of biomarkers into trial design for aGVHD offers an important means of defining eligibility criteria and streamlining patient selection. The MAGIC Algorithm Probability (MAP), which has been extensively validated in multicenter studies, integrates serum levels of ST2 and REG3 α to stratify patients

into low-, intermediate-, and high-risk groups that predict NRM and treatment response [64,65]. When measured at diagnosis, the MAP can be used as a tool to design risk-adapted approaches to treatment. For example, a recent phase 2 trial investigated the addition of natalizumab to corticosteroids for patients with high-risk aGVHD, while another phase 2 trial demonstrated that itactinib monotherapy in patients with low-risk aGVHD could be a compelling alternative to standard corticosteroid therapy [26,42]. However, an open question is whether the use of biomarkers for recruitment and inclusion retains the same efficiency and relevance in large-scale phase 3 trials. Moreover, the potential incorporation of biomarkers into response criteria could meaningfully transform trial methodology.

Adaptive trial design

Adaptive clinical trial designs hold particular promise in aGVHD due to the small patient population and need for rapid assessment of therapeutic efficacy. Such designs allow for pre-specified protocol modifications based on interim analyses, such as early stopping for futility or efficacy or addition of new treatment arms for promising agents. Multi-arm, multi-stage and adaptive trials can test several therapies in parallel, dropping inferior arms early and reallocating resources to more promising interventions [66]. This improved flexibility over traditional fixed designs, while statistically complex, accelerates identification of effective treatments as well as minimizing patient exposure to possibly ineffective therapies. However, implementation of adaptive designs would require substantial operational and financial resources. As such, an adaptive trial would benefit from coordination between major consortia, as well as pharmaceutical companies responsible for supplying

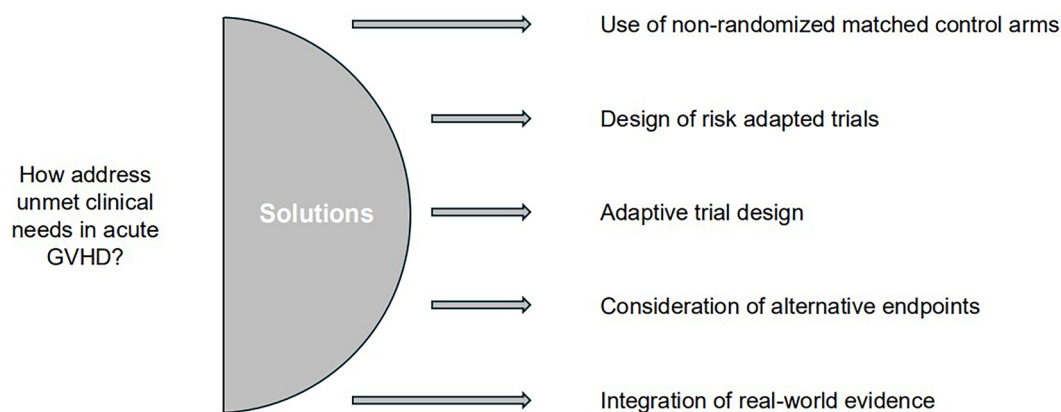


Figure 1. Approaches to address current limitations to clinical trial design in acute GVHD.

investigational agents and funding for their respective arms. Critics may argue, and reasonably so, that rigorously curated historical controls such as those from the MAGIC database may serve as a more practical and cost-effective comparison arm. Ultimately, while adaptive trial design represents an innovative and worthy avenue for advancing aGVHD research, its widespread application may remain constrained by logistical limitations.

Alternative endpoints for streamlined determination of clinical efficacy

ORR at day 28 remains the most widely accepted primary endpoint in aGVHD clinical trials, supported by its strong association with non-relapse mortality and long-term outcomes. However, other metrics may be more useful as primary endpoints in settings with small sample sizes or the need for rapid assessment of clinical efficacy to drive treatment decisions. Recent evidence has supported the use of earlier endpoints, such as day 14 ORR, in providing a reasonable assessment of clinical efficacy, which may also be a more pragmatic and efficient measurement in smaller trials [63]. Additionally, use of Day 28 CR (rather than ORR) as the primary endpoint in aGVHD trials, especially for patients with SR-aGVHD, may represent another approach to trial design. In a comparative analysis of ruxolitinib use for SR-aGVHD, a landmark analysis observed that achieving a CR by Day 28 (regardless of therapeutic agent) was most influential on risk for future NRM [23]. Designing trials to demonstrate improvements in CR rates may allow for smaller sample sizes while maintaining clinical impact.

Integration of real-world evidence

Integration of real-world evidence (RWE) is another innovative approach to aGVHD trial design. RWE allows researchers to incorporate data from patient registries and observational studies to supplement clinical trial findings, potentially providing a more comprehensive understanding of treatment effectiveness in diverse populations. One approach in this area is the creation of digital twin cohorts, which use RWE to simulate a control group based on data from patients with similar characteristics. While RWE and digital twins offer promising avenues for improving the efficiency and generalizability of clinical trials, their integration into regulatory decision-making remains limited, and their application has so far been limited to the study of chronic GVHD [67].

Conclusion

Despite recent therapeutic advances, aGVHD continues to face significant challenges in both clinical management and research innovation. While the introduction of ruxolitinib has transformed the treatment paradigm, existing therapeutics remain insufficient for many patients with high-risk disease. Given the decreasing number of severe cases due to enhanced GVHD prophylaxis, clinical development strategies that prioritize early intervention, or leverage novel strategies such as algorithmic risk stratification and real-world evidence-integrated designs offer a path forward despite operational barriers. Basic science-driven advances in immunology, microbiology, and computational modeling can be harnessed to guide precision therapies and optimized trial frameworks. As new interventions informed by these fields emerge, integration with rigorous yet pragmatic clinical trial designs will be essential to improving outcomes for patients.

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